C-SW312 – Healthcare Management System
Software Requirements Specification (SRS)
Based on the Volere Requirements Specification Template v16

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1. Project Drivers

1.1 Purpose of the Project

The purpose of this project is to specify and develop a Healthcare Management System for a medium-sized hospital or clinic. The system supports the end-to-end lifecycle of outpatient care, including patient registration, schedule and appointment management, clinical documentation, billing and payments, reminders, and operational reporting.

This Software Requirements Specification (SRS) defines the functional and non-functional requirements of the Healthcare Management System. It is based on the Volere Requirements Specification Template v16 and the analytical work from Deliverable 2 (Noun Technique, class diagram, CRUD analysis, state machines, activity diagrams, and System Sequence Diagrams). The SRS serves as a contract between stakeholders (teaching staff, student team, and the notional hospital client) and the development team, and as a reference for design, implementation, testing, and grading.

1.2 User Business / Background

The target environment is a hospital or outpatient clinic that manages patient care using a combination of manual processes and fragmented software. Core activities such as registering patients, booking and cancelling appointments, completing treatments, generating bills, sending reminders, and producing management reports are often handled in separate spreadsheets, paper forms, or ad hoc tools. This leads to:

- Scheduling conflicts and double-booked resources.
- Missed or late appointments due to lack of reminders.
- Delayed or inconsistent billing and payment tracking.
- Incomplete medical records and weak audit trails.
- Limited visibility into operational performance (e.g., appointment utilization, overdue bills).

The Healthcare Management System is intended as an integrated solution that:

- Centralizes core information about Patients, Doctors, Appointments, MedicalRecords, Prescriptions, Rooms, and Bills (as identified by the Noun Technique and class diagram).
- Coordinates the work of Patients, Receptionists, Doctors, Administrators, and automated schedulers around a consistent domain model.
- Automates recurring and time-dependent behaviours captured in the state machines, activity diagrams, and SSDs (e.g., appointment reminders, cleanup of expired slots, monthly billing).
- Provides a realistic case study for applying requirements engineering techniques (Volere template, atomic requirements with fit criteria, UML modeling, CRUD analysis) within the C-SW312 course.

1.3 Goals of the Project

G1 – Improve appointment and schedule management
The system shall provide reliable booking, confirmation, and cancellation workflows that prevent double-booking and enforce the hospital's cancellation policy.

- **Advantage:** Fewer conflicts and no-shows; more efficient use of doctors and rooms.
- **Measurement:** For all modeled scenarios (UC-01 to UC-12 and UC-T1–T3), appointments respect doctor and room availability, no overlapping confirmed appointments exist for the same doctor-room-timeslot combination, and the Appointment state machine transitions match the defined lifecycle (Requested, Active, Confirmed, InProgress, Completed, Cancelled, Expired).

G2 – Support accurate billing and payment handling
The system shall generate and manage Bills for completed treatments and support robust payment workflows, including failure and retry handling.

- **Advantage:** Clear financial records per patient, alignment between clinical activity and billing, and realistic integration with external payment services.
- **Measurement:** For the test scenarios, every completed treatment with billable items results in at least one corresponding Bill; each Bill progresses through the Bill lifecycle states (Created, PendingPayment, Paid, Overdue, WrittenOff) according to the specified business rules and use cases (UC-05, UC-07, UC-11, UC-T1).

G3 – Enhance information availability and auditability
The system shall maintain a single, consistent MedicalRecord per patient and trace key operations across use cases, requirements, and domain classes.

- **Advantage:** Safer clinical decision making, less duplication, and strong justification for design choices across Deliverables 2 and 3.

- **Measurement:** For each patient there is exactly one MedicalRecord aggregating history; the traceability matrix links each functional requirement (FR-xx) to at least one use case and one domain class; audit logs exist for critical operations such as registration, bookings, cancellations, discharges, payments, overdue and write-off changes.

G4 – Serve as a teaching exemplar for requirements engineering
The project shall demonstrate the application of the Volere Template and related techniques in a coherent, testable SRS.

- **Advantage:** Supports grading against clear, complete, and measurable requirements while aligning with the course learning outcomes.
- **Measurement:** The SRS covers all required sections of the adapted Volere structure for the course; each FR-xx and NFR-xx includes a fit criterion; all major use cases from Deliverable 2 are represented in the functional requirements and in the requirements–use case–domain class traceability matrix.

1.4 Stakeholders

Client / Sponsor (Course Perspective)

- **Role:** Course instructor(s) and teaching team for C-SW312 – Introduction to Software Engineering.
- **Interest:** Use the Healthcare Management System as an assessment artifact demonstrating correct application of requirements engineering techniques, UML modeling, and documentation quality.

Customer (Hospital Management)

- **Role:** Representative of hospital administration (e.g., Hospital Manager, Head of Outpatient Services).
- **Interest:** Ensure the system supports efficient day-to-day operations, enforces hospital policies (e.g., appointment cancellation rules, billing and write-off policies), and provides management information via reports.

Hands-On Users

- **Patients**
    - Book, view, and cancel appointments.
    - View and pay bills.

    - Receive notifications and reminders for upcoming appointments and outstanding payments.
- **Receptionists**
    - Register and update patient information.
    - Book, confirm, and cancel appointments on behalf of patients.
    - Help synchronize patient data with external systems where needed.
- **Doctors**
    - View and manage their schedules.
    - Complete patient treatments and record clinical notes.
    - Issue prescriptions and authorize discharges for admitted patients.
- **System Scheduler / Administrator**
    - Configure and trigger background jobs such as sending appointment reminders, generating weekly reports, monthly billing, and cleaning up expired slots.
    - Maintain system parameters and operational calendars.
- **Finance / Billing Staff**
    - Monitor bills and payment statuses.
    - Handle overdue bills, penalties, and write-off decisions in line with business rules.
    - Reconcile payments reported by the payment gateway.
- **IT Operations and Support**
    - Deploy, monitor, and maintain the system in the target environment.
    - Manage integrations with external services (Payment Gateway, NotificationService, ReportingService, ExternalPatientSystem).
    - Support backup, recovery, and configuration management.

External Systems and Services

- **Payment Gateway**
    - Processes electronic payments started from the Make payment use case; returns success or failure for each attempt.
- **NotificationService**

    - Delivers reminders and notifications (e.g., SMS/email) for bookings, cancellations, and upcoming appointments.
- **ReportingService**
    - Generates scheduled weekly and monthly operational reports based on data from the Healthcare Management System.
- **ExternalPatientSystem**
    - Provides or receives patient demographic and clinical data for synchronization scenarios.

Development Team (Student Team)

- **Members:** Adham Sobhy, Andrew Emad, Yassin Farrag, Ahmed.
- **Roles:** Act as requirements engineers, analysts, and designers for this case study.
- **Interest:** Use this SRS, together with the Deliverable 2 models (Noun Technique tables, class diagrams, state machines, activity diagrams, SSDs), as the authoritative basis for any implementation or prototype and as evidence for grading.

2. Project Constraints

2.1 Mandated Constraints (technology, deadlines, budget)

C-01 – Core domain data must be persistent
The system shall persist all core domain concepts – Patient, Doctor, Appointment, MedicalRecord, Prescription, Room, Bill – in a durable data store so that clinical and financial history can be retrieved over time.

C-02 – No physical deletion of MedicalRecord data
The system shall not physically delete MedicalRecord data. Logical deactivation or archival is allowed only through defined maintenance or administrative procedures, and historical clinical information must remain accessible for audit and safety.

C-03 – Controlled deletion of other entities via maintenance use cases
Deletion or cleanup of Appointments, slots, and related transient data shall be performed via explicit maintenance use cases (e.g., Clean up expired slots) rather than ad hoc bulk deletion, and shall preserve an audit trail.

C-04 – External payment gateway integration
All electronic payments shall be handled through an external secure Payment Gateway. The Healthcare Management System shall not store full payment card details, but only non-sensitive identifiers and status returned by the gateway.

C-05 – Course deadline constraint
The SRS, diagrams, and any prototype produced from it shall be completed within the schedule defined for the C-SW312 course deliverables. Scope and level of detail must be achievable within this deadline while still satisfying the functional and non-functional requirements documented here.

C-06 – Educational budget constraint
The project is conducted as a course assignment with no real monetary budget. The solution must assume the use of standard open-source or institutionally provided tools and platforms, without dependence on expensive commercial licenses beyond what is already available in the teaching environment.

2.2 Relevant Facts and Assumptions

The following facts and high-level assumptions shape the requirements and design of the Healthcare Management System:

- Each patient is uniquely identified within the system (e.g., by patientID).
- There is exactly one long-lived MedicalRecord per patient, aggregating all consultations and clinical notes.
- An Appointment links exactly one Patient and one Doctor at a specific date and time, and may reserve a Room.
- Bills are generated from completed treatments and track payment status, overdue handling, and possible write-offs.
- External services (Payment Gateway, NotificationService, ReportingService, ExternalPatientSystem) provide stable APIs for integration.
- More detailed assumptions and dependencies are described in Section 7 of this SRS.

2.3 Implementation Environment and Partner Systems

The implementation environment describes the technical context in which the Healthcare Management System operates within a medium-sized hospital or clinic. The system assumes:

- A persistent data store (e.g., a relational database) able to store core domain classes such as Patient, Doctor, Appointment, MedicalRecord, Prescription, Room, and Bill.
- One or more application servers hosting the business logic and APIs that implement the functional and non-functional requirements in this SRS.
- User access via standard desktop or web clients for receptionists, doctors, administrators, finance staff, and IT operations.

- The system collaborates with the following external or partner applications at the system boundary:
- Payment Gateway: Processes electronic payments for bills initiated from the "Make payment" use case and returns success or failure responses that drive the Bill lifecycle (PendingPayment, Paid, Overdue, WrittenOff).
- NotificationService: Delivers SMS/email notifications for reminders, confirmations, and cancellations, and reports delivery success or failure for logging and summary reporting.
- ReportingService: Generates weekly and monthly operational reports requested by the system scheduler and returns summaries or report artifacts.
- ExternalPatientSystem: Supplies or receives patient demographic and clinical data when synchronizing patient information and is invoked by the "Synchronize patient information" use case and related SSD.

These partner systems are treated as black boxes; only their interfaces and observable behaviours (as shown in the SSDs) are relevant.

2.4 Naming Conventions and Terminology

The Healthcare Management System uses consistent naming conventions across this SRS and the supporting UML models:

- Domain classes use UpperCamelCase names: Patient, Doctor, Appointment, MedicalRecord, Prescription, Room, Bill, Consultation, Person, Staff, Nurse, Medicine.
- Use cases are identified by IDs and descriptive names such as:
  UC-01 Book appointment
  UC-02 Register patient information
  UC-03 Manage doctor schedule

UC-04 Confirm appointment slot
UC-05 Complete patient treatments
UC-06 Cancel appointment
UC-07 Make payment
UC-08 Authorize patient discharge
UC-09 Send appointment reminders
UC-10 Generate weekly report schedule
UC-11 Generate monthly bill
UC-12 Clean up expired slots
UC-T1 Handle payment failure
UC-T2 Update slot availability
UC-T3 Synchronize patient information

- Functional requirements are labeled FR-xx; non-functional requirements are labeled NFR-xx.

- Diagrams use the same identifiers as the related use cases and domain classes (e.g., "State Machine #1 – Bill Lifecycle," "Activity Diagram – UC-01 Book appointment," "SSD_UC01_BookAppointment_Primary").

- Common acronyms include SRS, SSD, FR, NFR.

The glossary in the Data and Domain Model section provides concise definitions for domain terms such as Appointment, Bill, MedicalRecord, Consultation, and Reminder.

3. Functional Requirements

3.1 Scope of the Work

The scope of the work covers the operational processes involved in managing outpatient care and related administrative tasks at a hospital or clinic. The following business activities are in scope:

- Registering and maintaining patient information.
- Managing doctor schedules and appointment slots.
- Booking, confirming, cancelling, and completing appointments.
- Recording treatments and updating patient MedicalRecords.

- Generating and managing bills for completed treatments.
- Handling patient payments, including payment failures and retries.
- Sending appointment reminders and related notifications.
- Generating weekly and monthly operational reports.
- Running maintenance and cleanup jobs for expired or inconsistent data.
- Synchronizing patient information with an external system when required.

These activities are captured in the use cases defined in Deliverable 2, including UC-01 to UC-12 and UC-T1 to UC-T3.

3.2 Scope of the Product

The Healthcare Management System is responsible for automating the above work within a defined system boundary. The product:

- Maintains the core domain objects: Patient, Doctor, Appointment, MedicalRecord, Prescription, Room, Bill.
- Provides user interfaces and APIs for Patients, Receptionists, Doctors, Administrators, and Finance/Billing staff.
- Communicates with external services (Payment Gateway, NotificationService, ReportingService, ExternalPatientSystem) through defined interfaces.
- Manages state transitions for Appointments and Bills as specified by the state machine diagrams.
- Executes scheduled jobs for reminders, reporting, billing, and cleanup.

The product explicitly does **not** cover low-level hardware details, detailed UI screen design, or deep analytics beyond the reports described in the functional requirements.

3.3 Functional Requirements

FR-01 – Register new patient

- **Description:** The system shall allow a receptionist to register a new patient by entering core demographic and contact details.
- **Source:** UC-02 Register patient information; Noun Technique (Patient).

- **Rationale:** New patients must be captured before any clinical or billing activities can take place.
- **Fit Criterion:** Given a receptionist with access rights, when they complete all mandatory registration fields and submit, the system creates a new Patient record with a unique patientID and displays a confirmation.

FR-02 – Maintain patient information

- **Description:** The system shall allow authorized users to update existing patient information (e.g., address, phone, insurance details) while preserving historical data where required.
- **Source:** UC-02 Register patient information; UC-T3 Synchronize patient information.
- **Rationale:** Patient data frequently changes and must remain accurate.
- **Fit Criterion:** Given an existing patient, when an authorized user edits and saves changes, the updated fields are stored and visible in subsequent views of the patient profile.

FR-03 – Create and update medical record

- **Description:** The system shall automatically create a MedicalRecord for each new patient and allow doctors to update it with diagnoses, allergies, and clinical notes after each consultation.
- **Source:** UC-02 Register patient information; UC-05 Complete patient treatments.
- **Rationale:** A single, longitudinal medical record per patient is required for safe and consistent care.
- **Fit Criterion:** For each patient, exactly one MedicalRecord exists; after a completed treatment, new clinical notes are appended and visible in the patient's history.

FR-04 – Manage doctor schedules

- **Description:** The system shall allow a doctor or authorized staff to view, add, modify, and block timeslots in a doctor's schedule.
- **Source:** UC-03 Manage doctor schedule.
- **Rationale:** Controlled schedule management is needed to avoid conflicts and support booking.
- **Fit Criterion:** When a doctor submits schedule changes with no conflicts, the updated slots are stored and are reflected in subsequent appointment searches.

FR-05 – Book appointment

- **Description:** The system shall allow a patient to book an appointment with a doctor by selecting a date, time, and available slot.
- **Source:** UC-01 Book appointment.
- **Rationale:** Booking appointments is a core service of the hospital system.
- **Fit Criterion:** Given an available slot and a valid patient, when booking is requested, an Appointment with status Requested or Confirmed is created and linked to the Patient and Doctor.

FR-06 – Confirm appointment slots

- **Description:** The system shall allow a receptionist or authorized staff to confirm pending appointments and assign them to a final room and timeslot.
- **Source:** UC-04 Confirm appointment slot.
- **Rationale:** Some organizations require manual confirmation before appointments become final.
- **Fit Criterion:** When a pending appointment is confirmed and the slot is still valid, the Appointment status changes to Confirmed and the assigned Room is recorded.

FR-07 – Prevent schedule conflicts

- **Description:** The system shall prevent double-booking of the same doctor, room, and timeslot.
- **Source:** UC-01 Book appointment; UC-03 Manage doctor schedule; UC-04 Confirm appointment slot; UC-T2 Update slot availability.
- **Rationale:** Avoiding conflicts is critical for operational efficiency and patient satisfaction.
- **Fit Criterion:** When a new appointment or schedule change would overlap an existing confirmed Appointment for the same doctor and room, the system rejects the change and explains the conflict.

FR-08 – Cancel appointment

- **Description:** The system shall allow a patient or receptionist to cancel a confirmed appointment, subject to the cancellation policy.
- **Source:** UC-06 Cancel appointment.
- **Rationale:** Patients may need to reschedule and the system must free slots accordingly.

- **Fit Criterion:** When a valid cancellation request is made more than 24 hours before the appointment time, the Appointment status is set to Cancelled, the slot is freed, and notifications are sent to both patient and doctor.

FR-09 – Handle late cancellation

- **Description:** The system shall enforce the cancellation policy for late cancellations (≤ 24 hours before the appointment) by either blocking cancellation or applying late-cancellation rules.
- **Source:** UC-06 Cancel appointment; Appointment State Machine.
- **Rationale:** Late cancellations should be treated differently for fairness and revenue protection.
- **Fit Criterion:** For cancellation requests within 24 hours of the appointment time, the system displays a warning and either refuses cancellation or records a CancelledLate status and logs any applicable penalty according to configuration.

FR-10 – Complete treatments and appointments

- **Description:** The system shall allow a doctor to mark an appointment as Completed and record treatment details at the end of a consultation.
- **Source:** UC-05 Complete patient treatments; Appointment State Machine.
- **Rationale:** Completion and documentation of treatment are required for clinical and billing processes.
- **Fit Criterion:** When the doctor submits valid treatment details for an in-progress appointment, the Appointment status becomes Completed, the MedicalRecord is updated, and optional Prescription data is stored.

FR-11 – Clean up expired appointments and slots

- **Description:** The system shall automatically detect and update appointments and slots that have expired based on configured rules (e.g., no-show or time window elapsed).
- **Source:** UC-12 Clean up expired slots; Appointment State Machine.
- **Rationale:** Expired or unused slots must be released to keep schedules consistent.
- **Fit Criterion:** When the cleanup job runs, appointments that satisfy the expiry conditions are updated to an Expired state and their associated slots are freed; a summary of affected records is produced.

FR-12 – Generate bills for completed treatments

- **Description:** The system shall generate Bill records for completed treatments and services, either at the time of completion or as part of a periodic billing run.
- **Source:** UC-05 Complete patient treatments; UC-11 Generate monthly bill; Bill State Machine.
- **Rationale:** Bills must be generated accurately to support payment and reporting.
- **Fit Criterion:** For each completed treatment with billable items, at least one Bill is created or updated with the correct patient, amount, and due date.

FR-13 – Monthly billing run

- **Description:** The system shall support a monthly billing process that consolidates relevant billable items into bills and updates their statuses.
- **Source:** UC-11 Generate monthly bill.
- **Rationale:** Many hospitals generate monthly statements for patients.
- **Fit Criterion:** When the monthly billing job runs for a target period, it reports how many bills were generated or updated and the total billed amount.

FR-14 – Make payment

- **Description:** The system shall allow a patient to pay outstanding bills via an external payment gateway.
- **Source:** UC-07 Make payment; UC-T1 Handle payment failure; Bill State Machine.
- **Rationale:** Electronic payments are required for efficient revenue collection.
- **Fit Criterion:** Given a bill in PendingPayment status, when the patient confirms a valid payment and the gateway returns success, the Bill status becomes Paid and the transaction is recorded.

FR-15 – Handle payment failures and retries

- **Description:** The system shall handle failed payment attempts by informing the patient, logging the failure, and allowing limited retries.
- **Source:** UC-T1 Handle payment failure; UC-07 Make payment; Appointment and Bill State Machines.
- **Rationale:** Robust error handling is required to deal with declined or timed-out transactions.

- **Fit Criterion:** When the payment gateway indicates failure, the system shows a clear error message and lets the patient either retry the payment (up to a configured maximum) or cancel, leaving the bill in PendingPayment.

FR-16 – Overdue and write-off handling

- **Description:** The system shall automatically mark bills as Overdue after their due date and allow authorized staff to mark long-overdue bills as WrittenOff.
- **Source:** Bill State Machine; billing policies.
- **Rationale:** Overdue tracking and write-offs are necessary for financial control.
- **Fit Criterion:** After the due date passes without full payment, the Bill status changes to Overdue; when an authorized user chooses to write off a bill, the status becomes WrittenOff and the change is logged.

FR-17 – Send appointment reminders

- **Description:** The system shall send reminders for upcoming appointments to patients and/or doctors according to configurable lead times.
- **Source:** UC-09 Send appointment reminders; Appointment State Machine.
- **Rationale:** Reminders reduce no-shows and improve utilization.
- **Fit Criterion:** When the reminder job runs, all appointments within the configured reminder window generate reminder messages; the system records which reminders were successfully sent.

FR-18 – Manage notification delivery via services

- **Description:** The system shall delegate delivery of reminders and notifications to an external NotificationService while tracking delivery success or failure.
- **Source:** UC-09 Send appointment reminders; SSDs.
- **Rationale:** Separation of concerns between core system and communication channels.
- **Fit Criterion:** For each reminder batch, the system logs the number of successfully delivered and failed notifications as reported by the NotificationService.

FR-19 – Generate weekly operational reports

- **Description:** The system shall generate weekly operational reports (e.g., appointments, cancellations, utilization) on request from a scheduler or administrator.
- **Source:** UC-10 Generate weekly report schedule.
- **Rationale:** Management needs regular visibility into system usage and performance.

- **Fit Criterion:** When the weekly report job runs, at least one report file or summary is produced, or the system explicitly reports that no data is available for the period.

FR-20 – Update slot availability

- **Description:** The system shall maintain up-to-date availability of appointment slots based on doctor schedules, bookings, cancellations, and maintenance jobs.
- **Source:** UC-T2 Update slot availability; UC-01 Book appointment; UC-03 Manage doctor schedule; UC-12 Clean up expired slots.
- **Rationale:** Slot availability must remain consistent across all booking and scheduling operations.
- **Fit Criterion:** After the update job runs, the number of free, reserved, and blocked slots matches the expected state derived from current appointments and schedules.

FR-21 – Synchronize patient information with external systems

- **Description:** The system shall synchronize selected patient data with an external patient information system when requested by authorized staff.
- **Source:** UC-T3 Synchronize patient information.
- **Rationale:** Some patient details originate from or must be shared with external systems.
- **Fit Criterion:** When synchronization is triggered for a patient, data differences are resolved according to configured rules; on success, local Patient and MedicalRecord data reflect the latest agreed information or an error is reported if synchronization fails.

FR-22 – Audit key operations

- **Description:** The system shall log key operations (e.g., registration, schedule changes, cancellations, payments, write-offs, discharges) with timestamps and user identifiers.
- **Source:** All use cases; legal and audit needs.
- **Rationale:** Auditing is required for safety, accountability, and compliance.
- **Fit Criterion:** For each audited operation, an entry exists in the audit log specifying the actor, timestamp, operation type, and affected entities.

4. Data and Domain Model

4.1 Reference to Domain Class Diagram

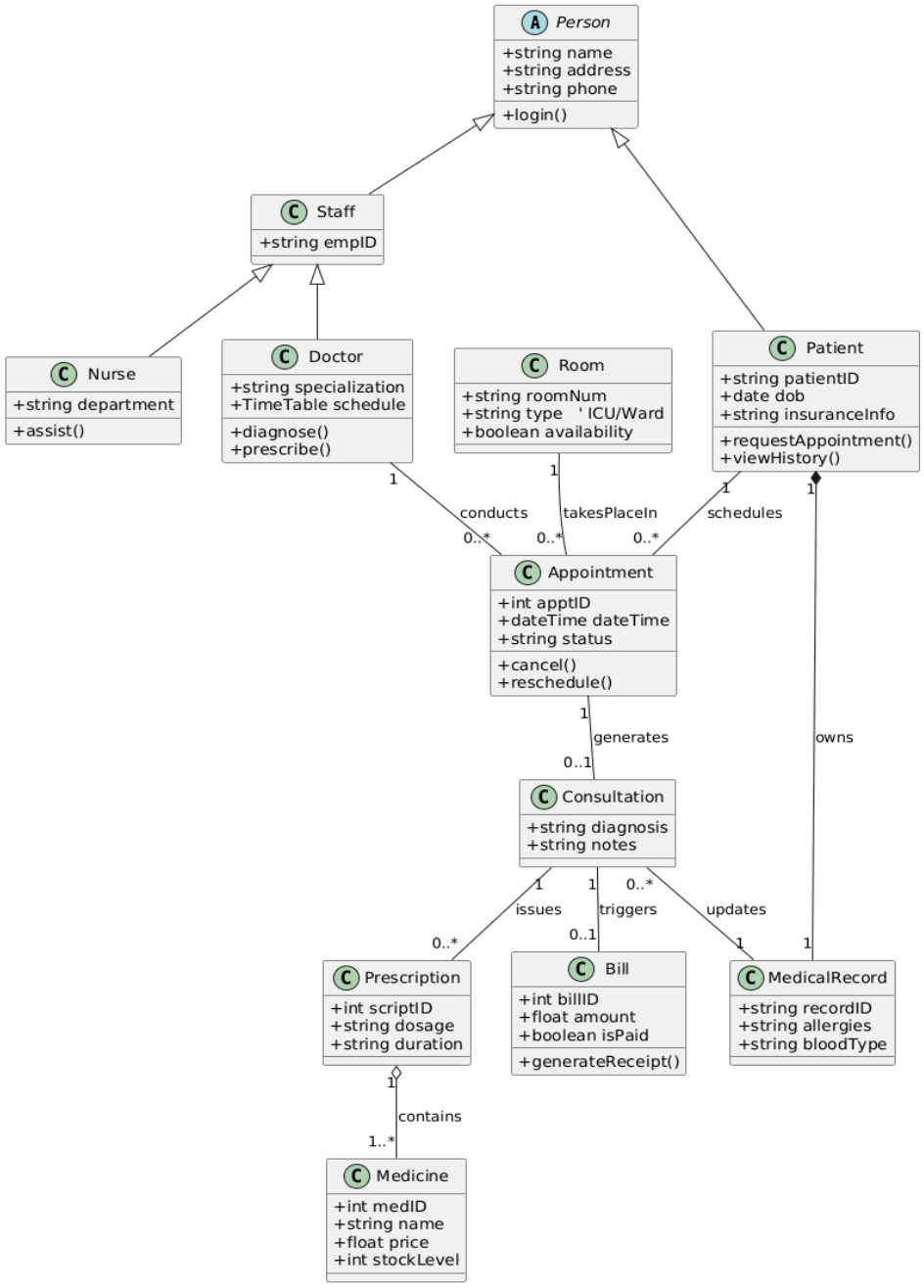
The Healthcare Management System is built around an agreed domain model derived from the Noun Technique and CRUD analysis in Deliverable 2. The seven core persistent domain classes are:

- Patient
- Doctor
- Appointment
- MedicalRecord
- Prescription
- Room
- Bill

Additional supporting classes such as Person, Staff, Nurse, Consultation, and Medicine are introduced in the UML Class Diagram as refinements or generalizations. The class diagram defines:

- Associations between patients, doctors, appointments, rooms, medical records, prescriptions, and bills.
- Multiplicities (e.g., one Patient has exactly one MedicalRecord; one Prescription can contain multiple Medicine items).
- Generalization hierarchies (e.g., Patient and Staff inherit from Person; Doctor and Nurse specialize Staff).
- Composition and aggregation relationships (e.g., Patient -- *MedicalRecord; Prescription o-- Medicine).*

This SRS assumes that the implementation will be consistent with that domain model.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conducts
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takesPlaceIn
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schedules
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owns
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+cancel()
+reschedule()
1
generates
0..1
Consultation
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1
issues
0..*
1
triggers
0..1
0..*
updates
1
Prescription
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+string dosage
+string duration
Bill
+int billID
+float amount
+boolean isPaid
+generateReceipt()
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+string recordID
+string allergies
+string bloodType
1
contains
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Medicine
+int medID
+string name
+float price
+int stockLevel

4.2 Glossary of Key Terms

Term	Definition
Patient	An individual receiving medical care and associated with exactly one MedicalRecord.
Doctor	A member of medical staff authorized to provide consultations and complete treatments.
Appointment	A scheduled meeting between a Patient and a Doctor at a specific date, time, and location.
MedicalRecord	The longitudinal clinical history for a Patient, including diagnoses, allergies, and notes.
Consultation	A clinical encounter between a Patient and a Doctor, typically associated with an Appointment.
Prescription	A set of instructions for medicines prescribed to a Patient after a consultation.
Room	A physical location (e.g., consultation room, ward bed) assigned to patients and appointments.
Bill	A financial record summarizing charges for treatments and services provided to a Patient.
Reminder	A notification sent to patients or doctors about upcoming appointments or outstanding bills.
System scheduler	An automated or administrative actor that initiates batch jobs (reminders, reports, billing, cleanup).
Payment Gateway	An external service that processes electronic payments initiated by the system.
NotificationService	An external service responsible for delivering SMS/email or other notifications.
ReportingService	An external service that generates operational reports from system data.
ExternalPatientSystem	An external system with which patient data may be synchronized.

4.3 Business Data Model and CRUD Overview

The business data model is derived from the Noun Technique, the domain class list, noun classification, and CRUD analysis from Deliverable 2. Core persistent domain classes include:

- Patient – individual receiving care.
- Doctor – medical staff conducting consultations.
- Appointment – scheduled meeting between a Patient and a Doctor.
- MedicalRecord – single longitudinal record per patient containing history and clinical notes.
- Prescription – details of medication ordered after treatments.
- Room – physical location assigned to patients and appointments.
- Bill – invoice generated for services rendered.

Supporting classes (Person, Staff, Nurse, Consultation, Medicine) refine and generalize these concepts. Key modeling decisions include:

- Use of an abstract Person class, with Patient and Staff as specializations and Staff further specialized into Doctor and Nurse.
- Composition between Patient and MedicalRecord, ensuring each patient has exactly one medical record.
- Aggregation between Prescription and Medicine, allowing medicines to exist independently as inventory items.
- Separation between scheduling (Appointment) and clinical work (Consultation, MedicalRecord).

CRUD responsibilities include:

- Patient and MedicalRecord creation and updates via UC-02 and UC-05; MedicalRecord is update-only.
- Appointment creation via UC-01, updates via UC-03 and UC-04, cancellation via UC-06, and cleanup via UC-12.
- Bill creation and updates in UC-05 and UC-11, and updates in UC-07 and UC-T1.
- Deletions centralized in maintenance and cleanup processes such as UC-12.

This structure ensures:

- Each persistent class has clear CRUD ownership.
- Use cases align with domain objects and UML diagrams.
- Requirements FR-01 to FR-22 are supported by a stable model.

5. Non-Functional Requirements

5.1 Look and Feel / Usability

NFR-01 – Learnability for receptionists

- **Description:** The system shall be easy for receptionists to learn for core tasks (registration, booking, cancellation).
- **Fit Criterion:** New receptionists with basic computer literacy shall be able to perform registration and simple booking tasks after at most 2 hours of guided training.

NFR-02 – Task efficiency for booking

- **Description:** The system shall minimize the number of steps required for common tasks.
- **Fit Criterion:** Completing a standard appointment booking (for a known patient and doctor) shall require no more than 7 major user interactions from start to confirmation.

5.2 Performance

NFR-03 – Response time for interactive operations

- **Description:** The system shall provide acceptable response times for interactive operations (e.g., booking, cancellation, viewing bills).
- **Fit Criterion:** For at least 95% of interactive requests under normal load (up to 50 concurrent users), the system shall respond within 3 seconds.

NFR-04 – Background job completion time

- **Description:** The system shall execute batch jobs (reminders, reports, monthly billing, cleanup) within a reasonable time window.
- **Fit Criterion:** Under normal data volumes for a medium-sized hospital, the nightly batch of reminder, billing, and cleanup jobs shall complete within 30 minutes.

5.3 Operational / Environmental

NFR-05 – System availability window

- **Description:** The system shall be available during defined operational hours.
- **Fit Criterion:** During scheduled operating hours (e.g., 07:00–22:00 local time), the system shall be available at least 99% of the time over a one-month period, excluding planned maintenance.

NFR-06 – Failure handling and recovery for external services

- **Description:** The system shall handle failures of external services (payment gateway, notification, reporting, external patient systems) without data loss or inconsistent states.
- **Fit Criterion:** For simulated outages of external services, all affected requests either complete successfully later (retry) or are rolled back with clear error messages; no partial payments or orphaned appointments remain.

5.4 Maintainability & Support

NFR-07 – Configurable business policies

- **Description:** The system shall allow key business policies (e.g., cancellation time window, maximum payment retries, reminder lead times) to be configured without code changes.
- **Fit Criterion:** Authorized administrators can change these policy parameters through configuration or administration screens and see the new values applied to subsequent operations.

NFR-08 – Logging for support

- **Description:** The system shall provide structured logging of errors and key events to support troubleshooting.
- **Fit Criterion:** For a sample of known error scenarios (e.g., payment failure, notification failure, invalid data), corresponding log entries exist with timestamps, user IDs (if applicable), and clear error messages.

5.5 Security

NFR-09 – User authentication

- **Description:** The system shall require secure authentication for all non-public functions.
- **Fit Criterion:** All access to patient, schedule, billing, and reporting functions is restricted to authenticated users with valid credentials; unauthenticated requests are rejected.

NFR-10 – Role-based access control

- **Description:** The system shall enforce role-based permissions (e.g., Patient, Receptionist, Doctor, Administrator, Finance).
- **Fit Criterion:** Attempts to perform operations beyond the user's role (for example, a patient trying to modify another patient's record, or a receptionist writing off a bill without authorization) are denied and logged.

NFR-11 – Data confidentiality in transit

- **Description:** The system shall protect sensitive data transmitted over networks (e.g., login credentials, medical and billing data).
- **Fit Criterion:** All application traffic that includes personal or medical data is transmitted over encrypted channels (e.g., HTTPS/TLS); plain-text access endpoints are disabled.

5.6 Legal / Regulatory

NFR-12 – Confidentiality of medical records

- **Description:** The system shall protect the confidentiality of medical records in line with healthcare privacy principles applicable to the notional hospital.
- **Fit Criterion:** Access to MedicalRecord details is restricted to authorized clinical roles; attempts from unauthorized roles are denied and recorded in the audit trail.

NFR-13 – Audit trail completeness

- **Description:** The system shall maintain an audit trail for key clinical and financial events.
- **Fit Criterion:** For a sample of at least 20 random operations (e.g., bookings, cancellations, discharges, payments, write-offs), corresponding audit log entries exist with correct actor, timestamp, and operation details.

6. Business Rules

BR-01 – Appointment cancellation policy

- Patients may cancel appointments without penalty only if the cancellation occurs more than 24 hours before the scheduled appointment time.
- Cancellations at or within 24 hours of the appointment are treated as late cancellations and may be blocked or recorded with penalties, according to hospital policy.

BR-02 – Single medical record per patient

- Each patient shall have exactly one MedicalRecord that aggregates all consultations and clinical notes; multiple separate medical records for the same individual are not allowed.

BR-03 – Bill overdue and write-off

- A bill becomes Overdue when its due date has passed without full payment.

- Overdue bills may later be WrittenOff by authorized staff based on hospital financial policy; write-offs must be logged and cannot be reversed without explicit administrative action.

BR-04 – Discharge preconditions

- A patient may be discharged only if all required treatments are completed, critical medical information is recorded, and no blocking conditions (e.g., mandatory observations) remain. This is enforced in the Authorize patient discharge use case.

BR-05 – Payment before certain services

- For services requiring prepayment, the associated Appointment shall not reach Confirmed or InProgress states until the corresponding Bill is in a Paid or acceptable PendingPayment state as defined by hospital policy.

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7. Assumptions and Dependencies

A-01 – Unique patient identification
Each patient has a unique identifier within the hospital system (e.g., patientID), which is used to link appointments, medical records, and bills.

A-02 – Network connectivity
The system operates in an environment with reliable network connectivity sufficient to reach external services (Payment Gateway, NotificationService, ReportingService, ExternalPatientSystem).

A-03 – Availability of external services
External services required by the system (e.g., payment gateway, notification, reporting, external patient system) provide documented and reasonably stable interfaces, and are maintained by their respective providers.

A-04 – Trained users
Receptionists, doctors, administrators, and finance staff receive basic training on the system's user interface and workflows before operating it in a live environment.

Dependencies:
The successful operation of the Healthcare Management System depends on:

- Deployment on suitable hardware and infrastructure (database, application server, network).
- Availability of integration endpoints and credentials for external services.

- Institutional policies that clearly define cancellation rules, billing practices, and write-off criteria.

8. Traceability

8.1 Requirements–Use Case Traceability Matrix

The following matrix links each functional requirement to its main source use case(s), primary domain classes, and related diagrams.

Req ID	Related Use Case(s)	Main Domain Classes	Related Diagrams
FR-01	UC-02 Register patient information	Patient, MedicalRecord	SSD UC-02; Class Diagram (Patient, MedicalRecord)
FR-02	UC-02 Register patient information; UC-T3 Synchronize info	Patient, MedicalRecord	SSD UC-02, UC-T3; Class Diagram
FR-03	UC-02 Register patient information; UC-05 Complete treatments	Patient, MedicalRecord	SSD UC-02, UC-05; Class Diagram
FR-04	UC-03 Manage doctor schedule	Doctor, Appointment, Room	SSD UC-03; Class Diagram
FR-05	UC-01 Book appointment	Patient, Doctor, Appointment, Room	Activity UC-01; SSD UC-01; Appointment State Machine
FR-06	UC-04 Confirm appointment slot	Appointment, Room	SSD UC-04; Appointment State Machine
FR-07	UC-01, UC-03, UC-04, UC-T2 Update slot availability	Doctor, Appointment, Room	SSD UC-01, UC-03, UC-04, UC-T2; Class Diagram
FR-08	UC-06 Cancel appointment	Appointment, Room	Activity UC-06; SSD UC-06; Appointment State Machine
FR-09	UC-06 Cancel appointment	Appointment	Activity UC-06; Appointment State Machine
FR-10	UC-05 Complete patient treatments	Appointment, MedicalRecord, Prescription	SSD UC-05; Class Diagram; Appointment State Machine

Req ID	Related Use Case(s)	Main Domain Classes	Related Diagrams
FR-11	UC-12 Clean up expired slots	Appointment, Room	SSD UC-12; Appointment State Machine
FR-12	UC-05 Complete treatments; UC-11 Generate monthly bill	Bill, Appointment, Patient	SSD UC-05, UC-11; Bill State Machine
FR-13	UC-11 Generate monthly bill	Bill	SSD UC-11; Bill State Machine
FR-14	UC-07 Make payment	Bill	Activity UC-07; SSD UC-07; Bill State Machine
FR-15	UC-T1 Handle payment failure; UC-07 Make payment	Bill	SSD UC-T1; Activity UC-07; Bill State Machine
FR-16	Billing policies; maintenance/administrative use cases	Bill	Bill State Machine
FR-17	UC-09 Send appointment reminders	Appointment, Patient, Doctor	SSD UC-09; Appointment State Machine
FR-18	UC-09 Send appointment reminders	Appointment	SSD UC-09; SSD with NotificationService
FR-19	UC-10 Generate weekly report schedule	Bill, Appointment, MedicalRecord	SSD UC-10; reporting-related diagrams (if provided)
FR-20	UC-T2 Update slot availability; UC-01; UC-03; UC-12	Appointment, Doctor, Room	SSD UC-T2; SSD UC-01, UC-03, UC-12; Class Diagram
FR-21	UC-T3 Synchronize patient information	Patient, MedicalRecord	SSD UC-T3
FR-22	All use cases (for audited operations)	All core domain classes (via logged operations)	State Machines; SSDs; Class Diagram

9. Project Issues

9.1 Open Issues

The following items remain open at the SRS level but do not block design or implementation:

- Exact user interface layout and screen design are unspecified; only behaviour and usability goals are defined.
- The technology stack (database vendor, framework, deployment platform) is not fixed.
- Hospital-specific parameters (penalties, thresholds, timing rules) are configurable rather than hard-coded.

9.2 Risks

Key risks include:

- R-01 – Inconsistency between SRS and UML models due to misinterpretations.
- R-02 – Underestimation of implementation effort for all use cases UC-01 to UC-12 and UC-T1 to UC-T3.
- R-03 – Simplified external services in prototypes, resulting in behaviour that differs from real-world integrations.

9.3 Waiting Room (Future Enhancements)

Future enhancements, not in scope for this deliverable, include:

- In-patient bed management, ward transfers, and admission workflows.
- Advanced analytics dashboards (no-show prediction, capacity planning, revenue analytics).
- A patient-facing mobile application with push notifications and self-service features.
- Integration with laboratory and imaging systems for results import.

9.4 Ideas for Solutions

Although this SRS focuses on requirements, the following design ideas are noted:

- Use a layered architecture separating presentation, application logic, and domain model.

- Implement scheduled jobs for reminders, reporting, billing, and cleanup.
- Use RESTful or message-based APIs for external integrations, consistent with SSDs.

9.5 User Documentation and Training

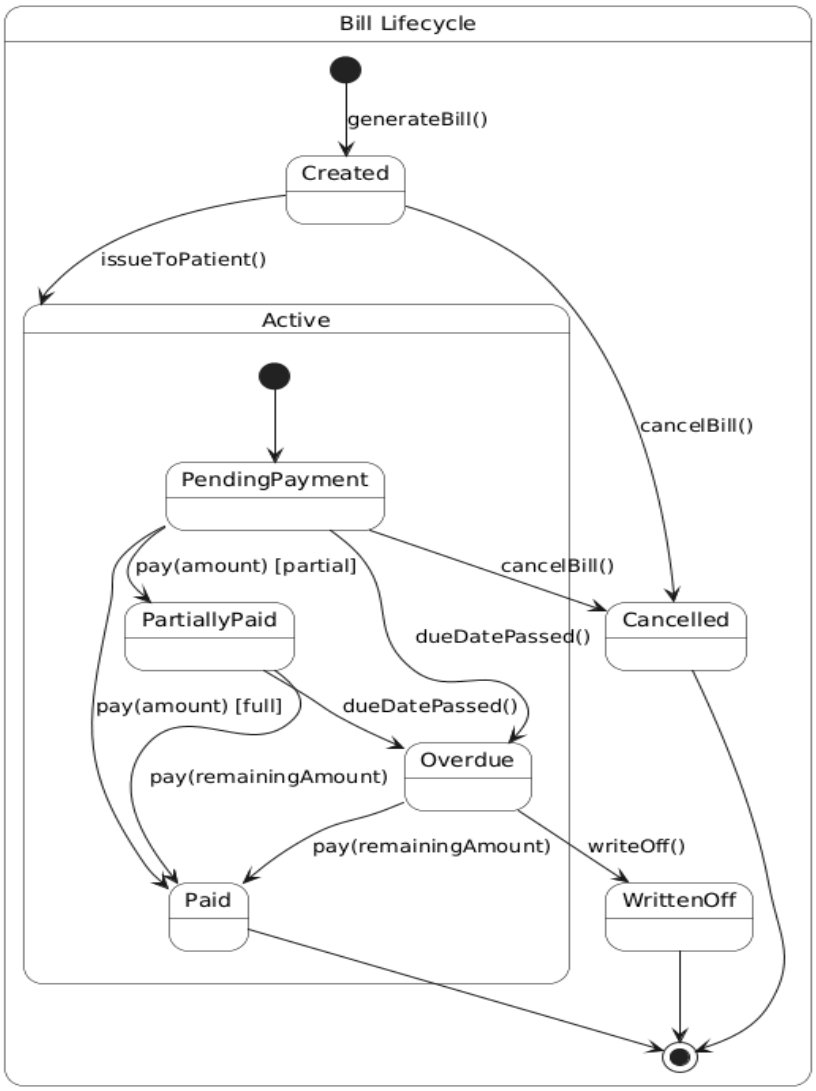
Expected documentation and training materials include:

- A user guide explaining tasks for Patients, Receptionists, Doctors, Administrators, and Finance staff.
- Short training sessions (up to 2 hours) for registration, booking, cancellation, treatments, and troubleshooting.
- Reference material for administrators and IT staff on configuration options and monitoring background jobs.

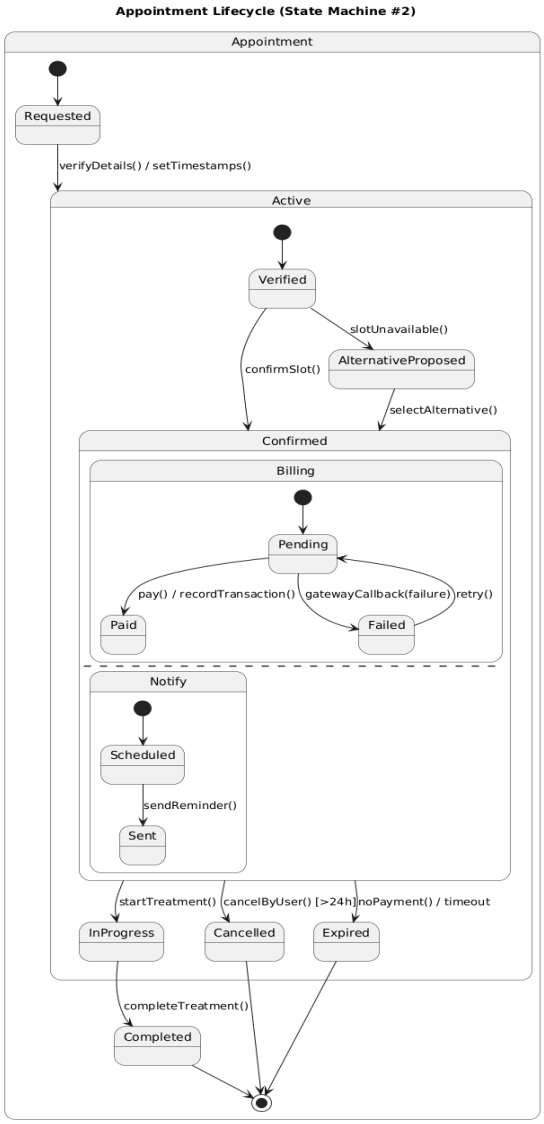
10. UML Diagrams and Models

10.1 State Machine Diagrams

State Machine #1 – Bill Lifecycle

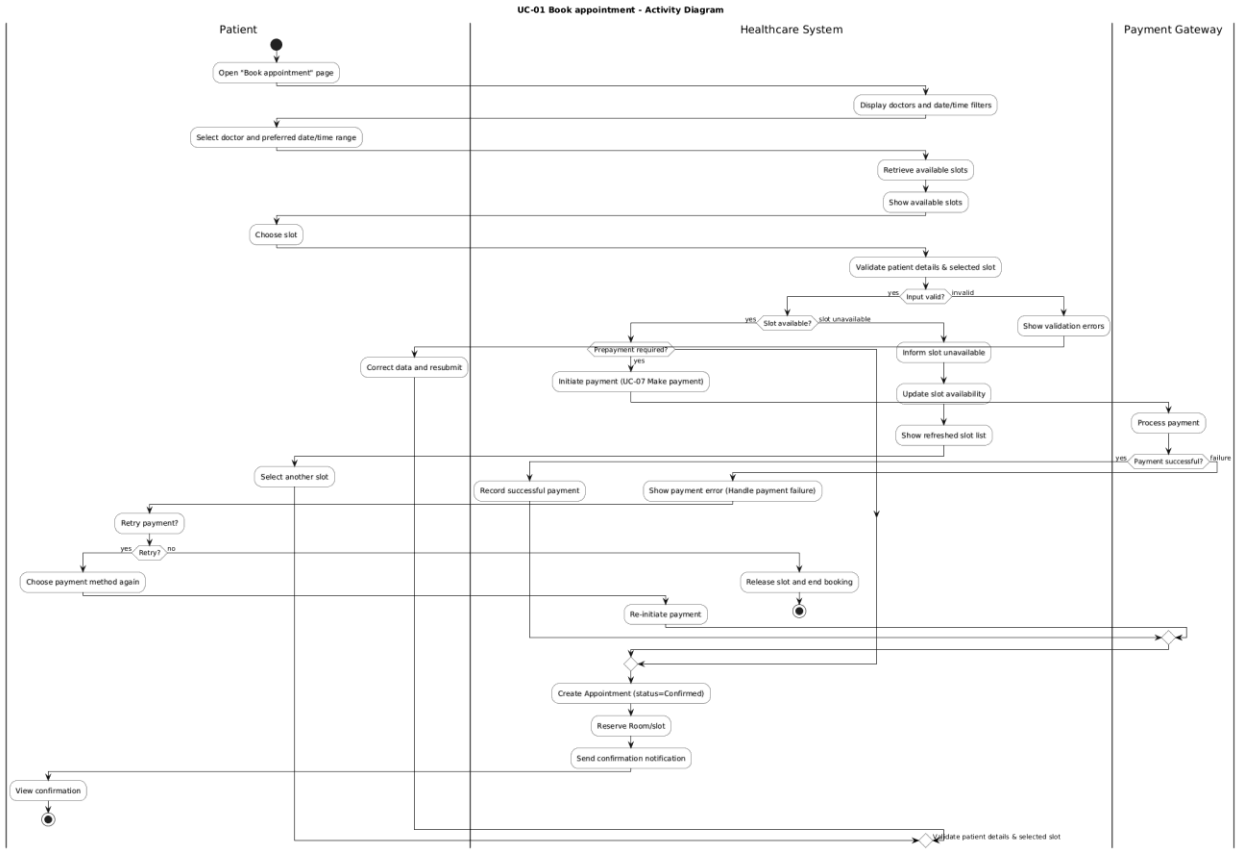


State Machine #2 – Appointment Lifecycle

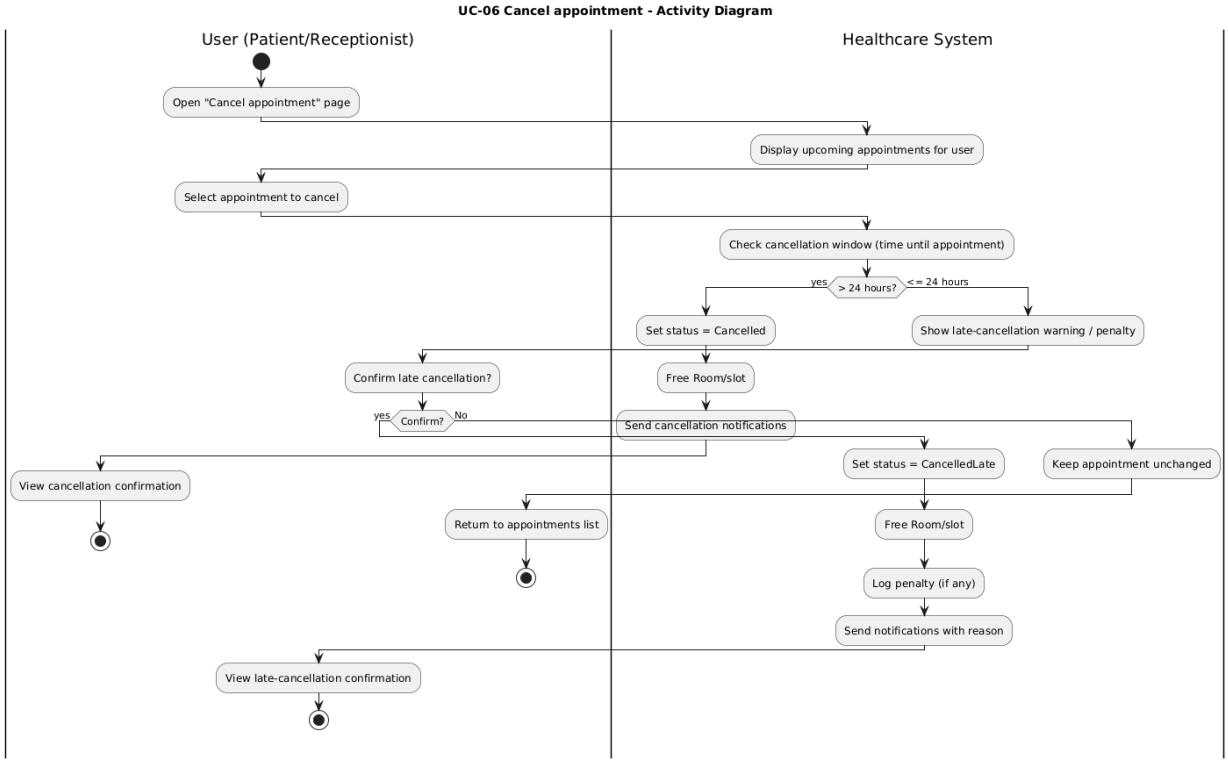


10.2 Activity Diagrams for Complex Use Cases

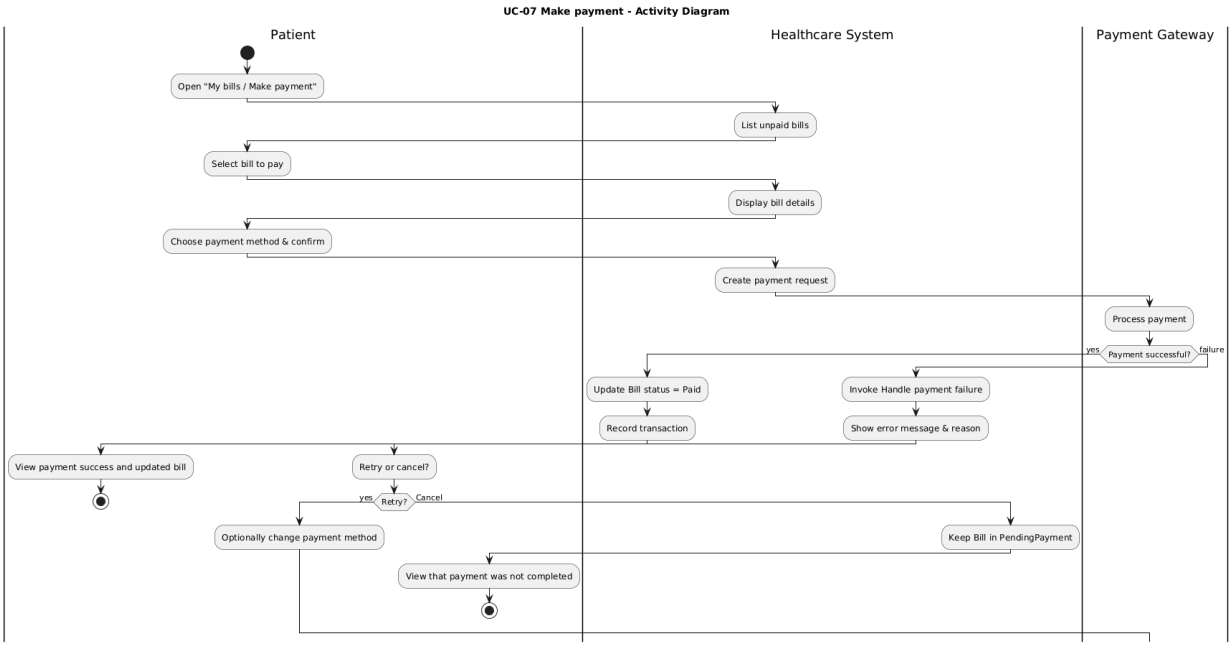
Activity Diagram – UC-01 Book appointment



Activity Diagram – UC-06 Cancel appointment

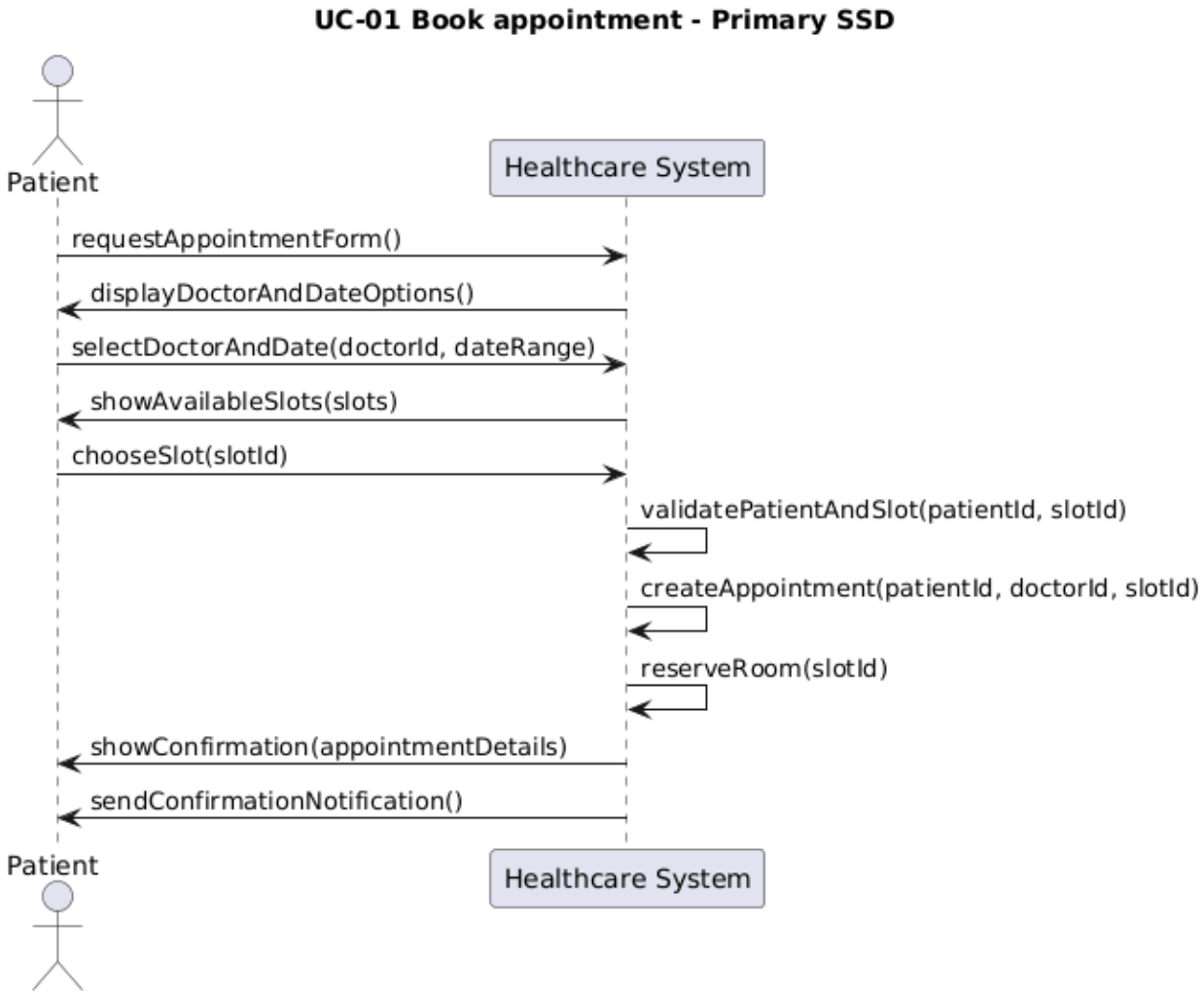


Activity Diagram – UC-07 Make payment

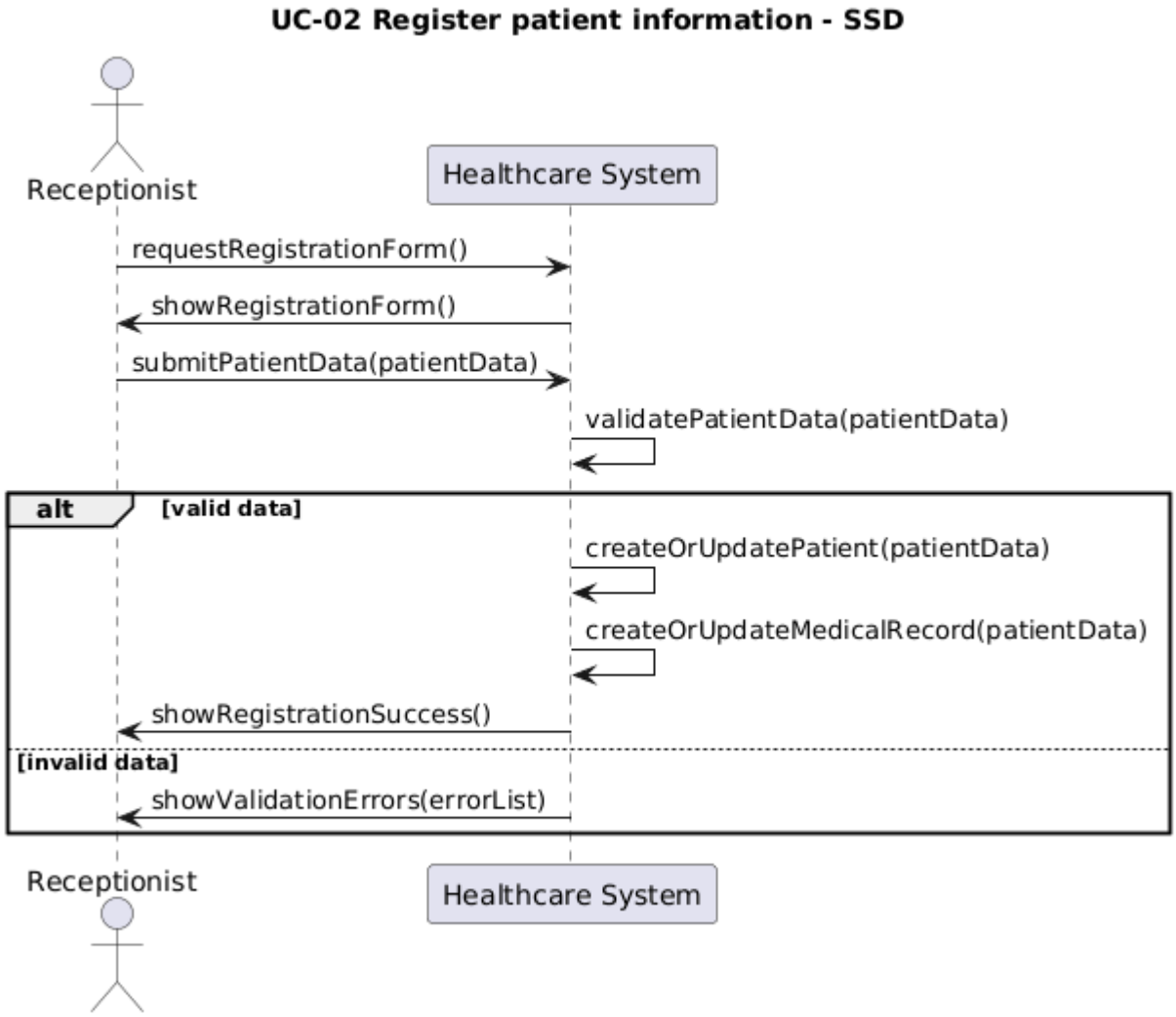


10.3 System Sequence Diagrams

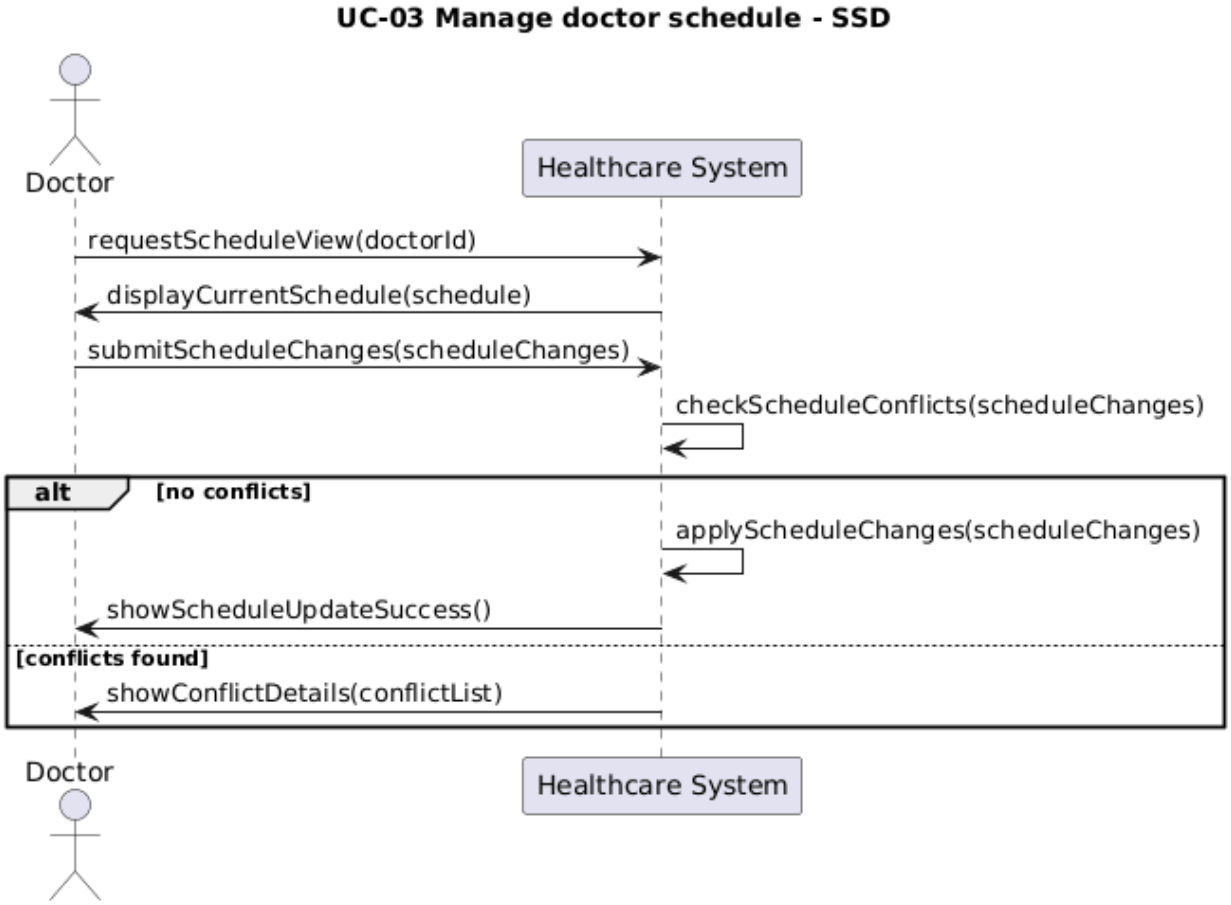
SSD – UC-01 Book appointment



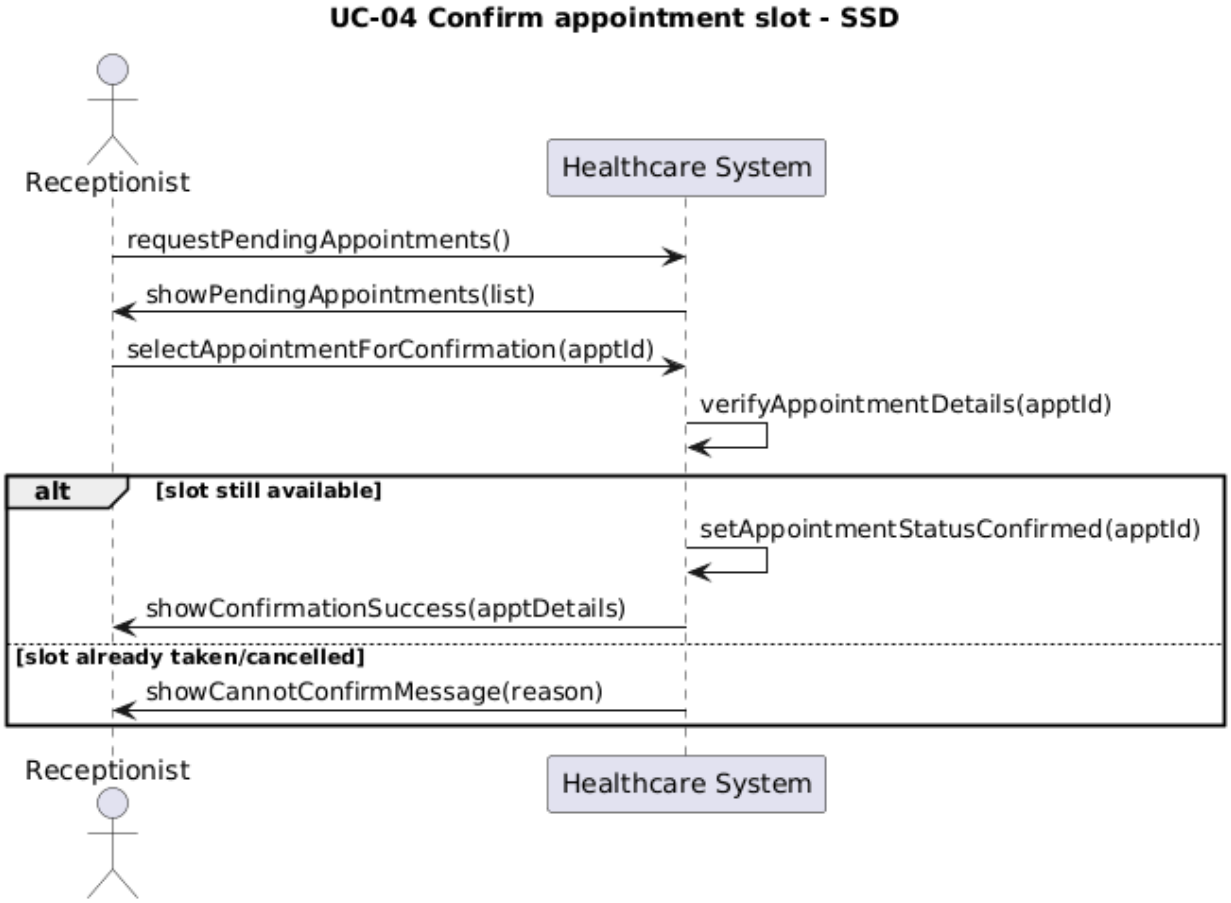
SSD – UC-02 Register patient information



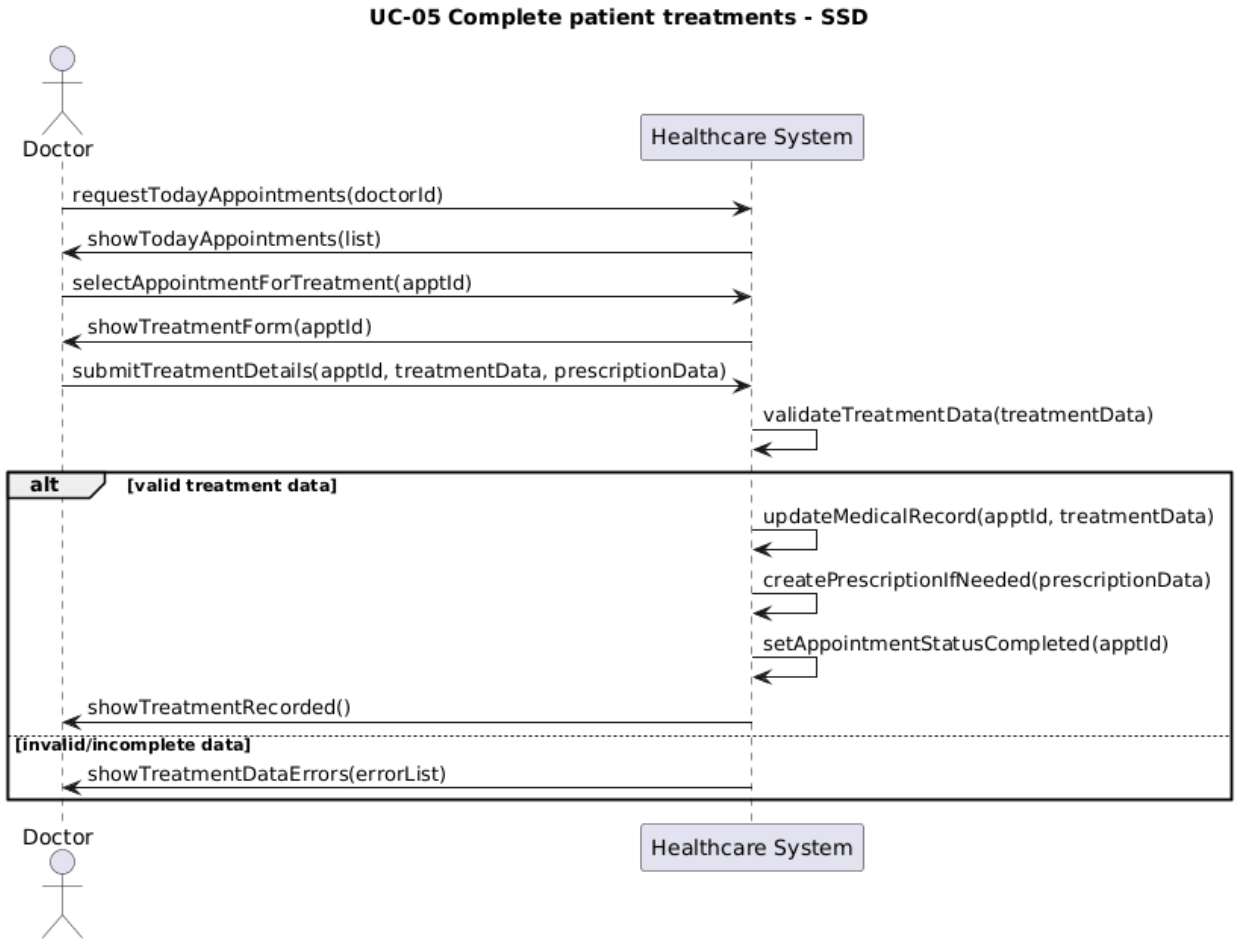
SSD – UC-03 Manage doctor schedule



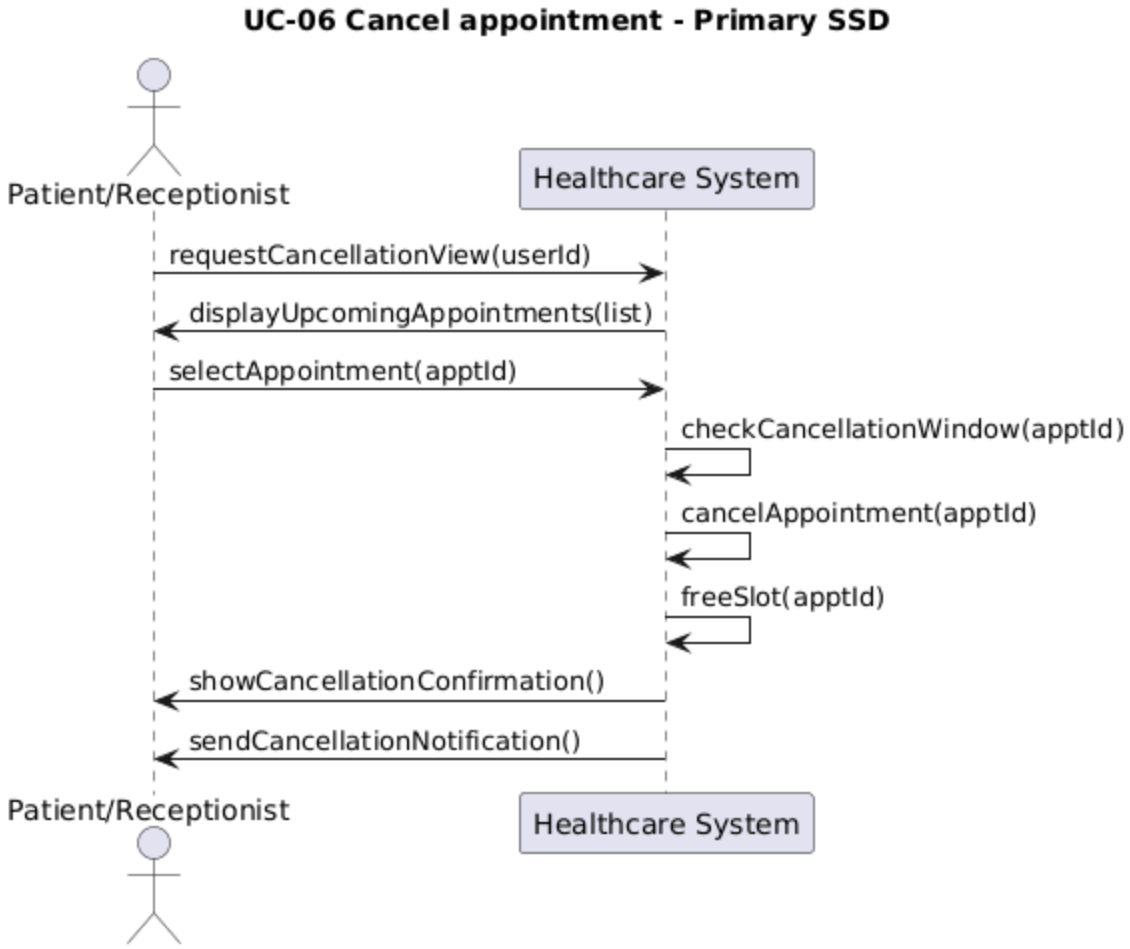
SSD – UC-04 Confirm appointment slot



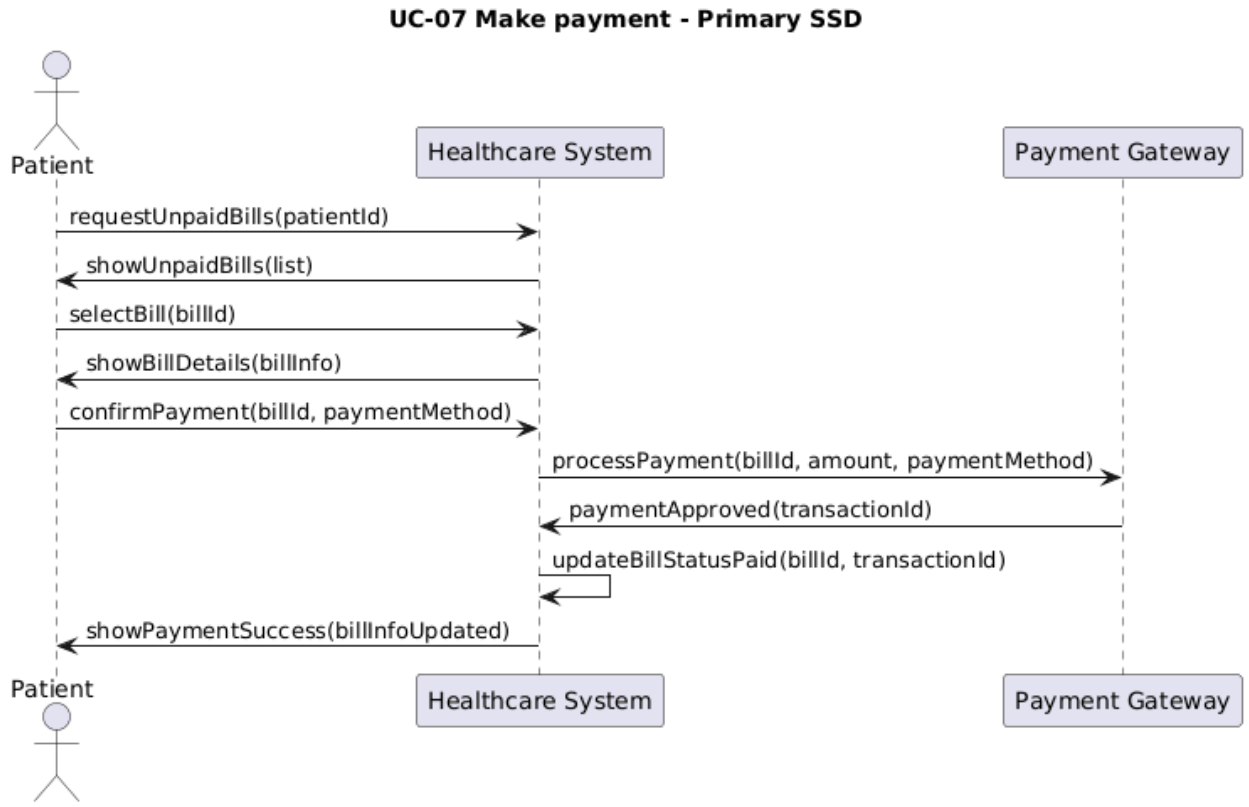
SSD – UC-05 Complete patient treatments



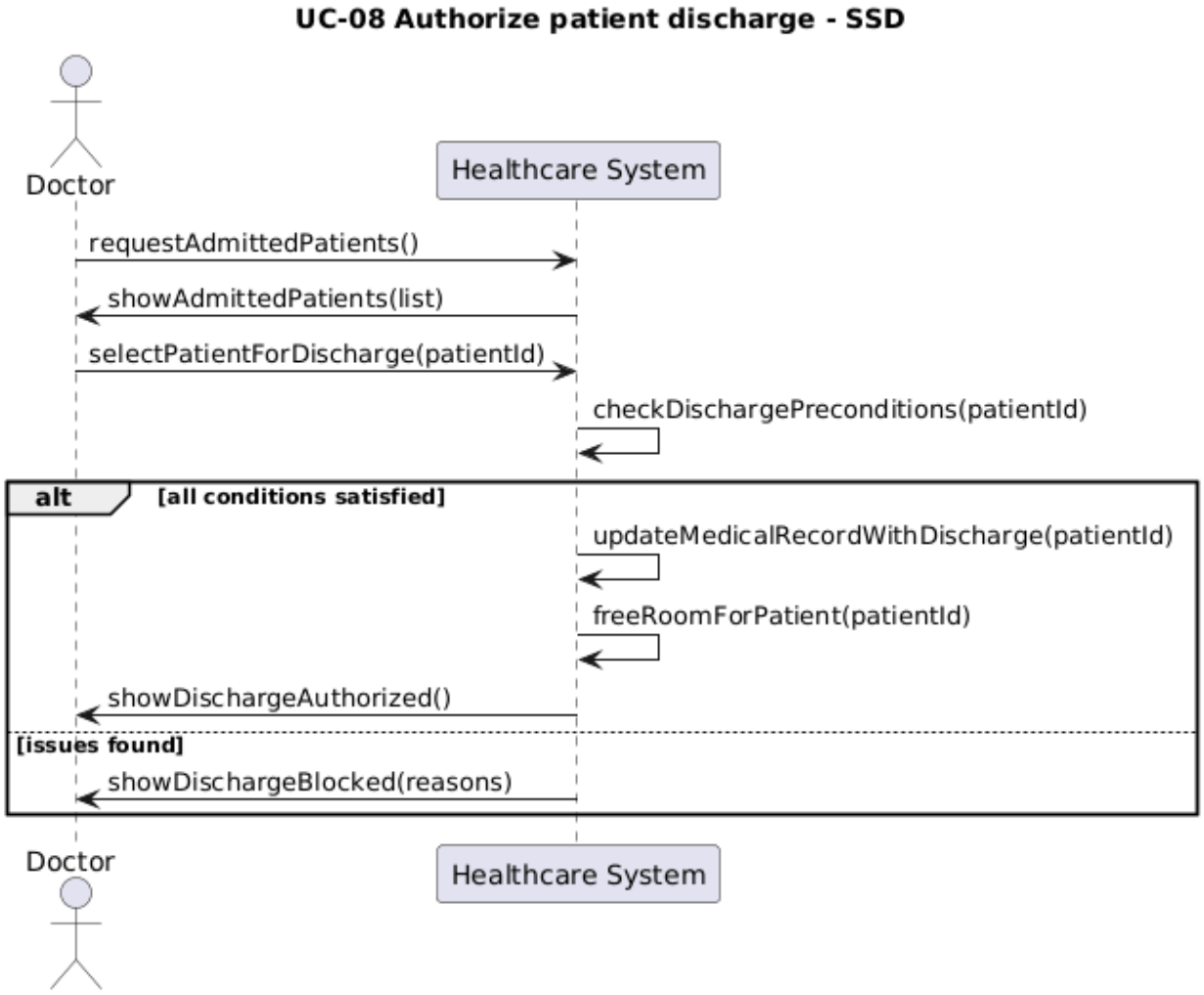
SSD – UC-06 Cancel appointment



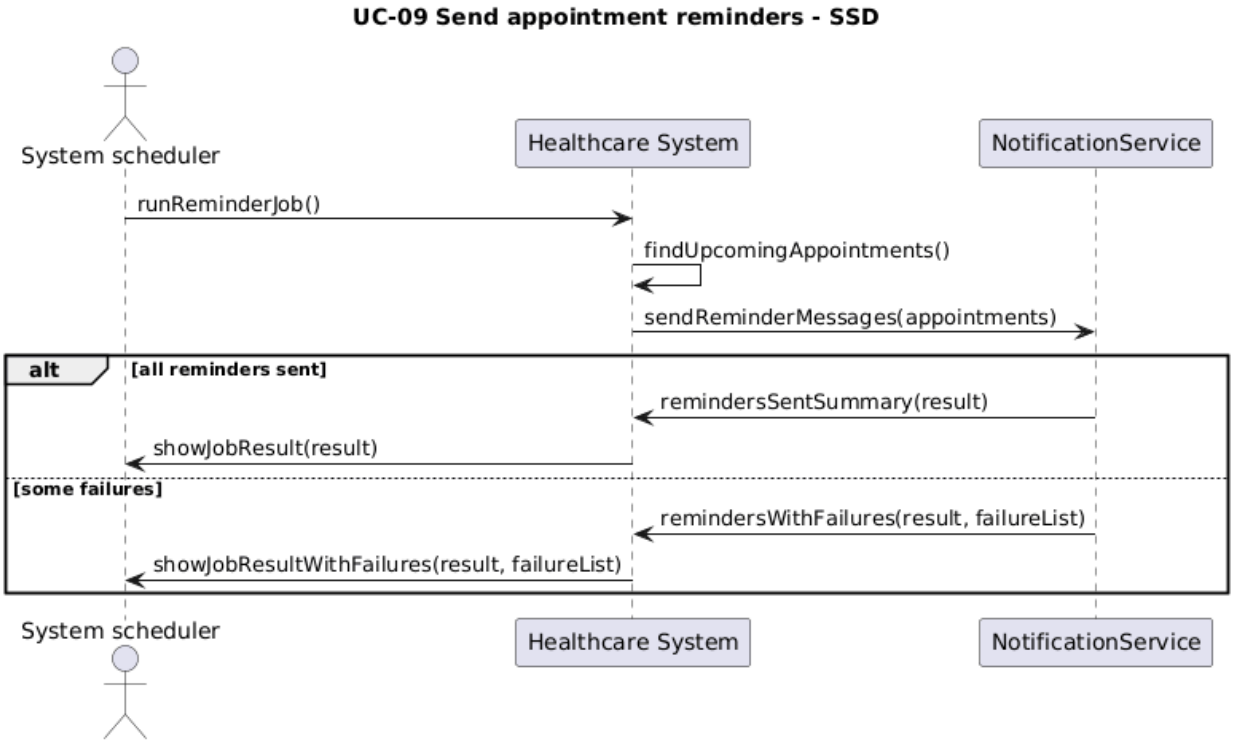
SSD – UC-07 Make payment



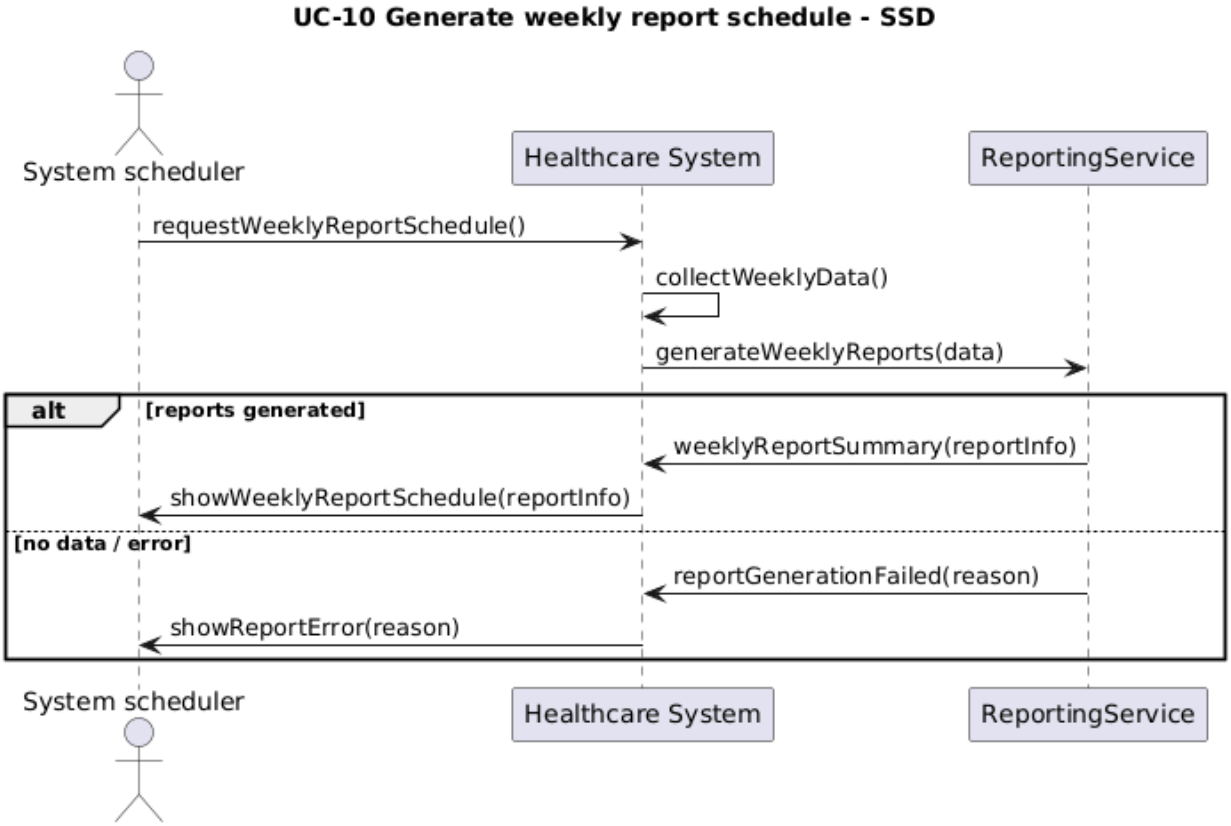
SSD – UC-08 Authorize patient discharge



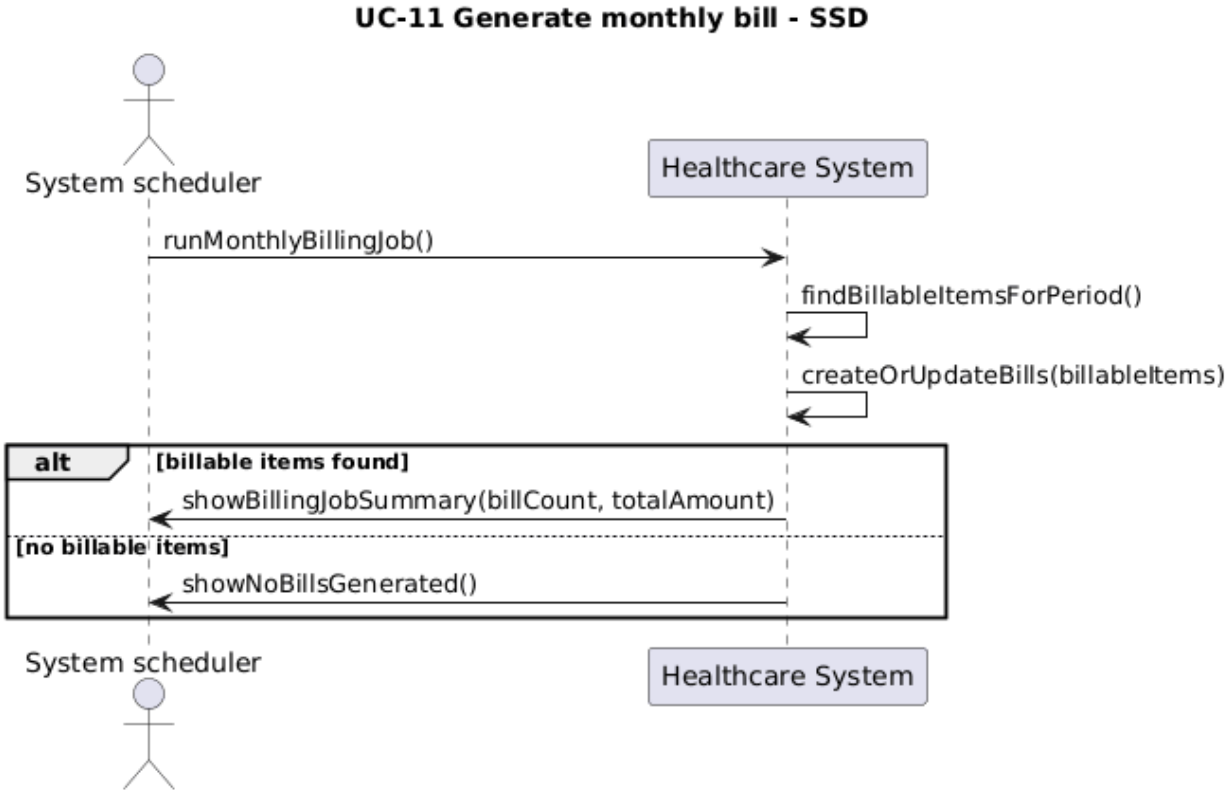
SSD – UC-09 Send appointment reminders



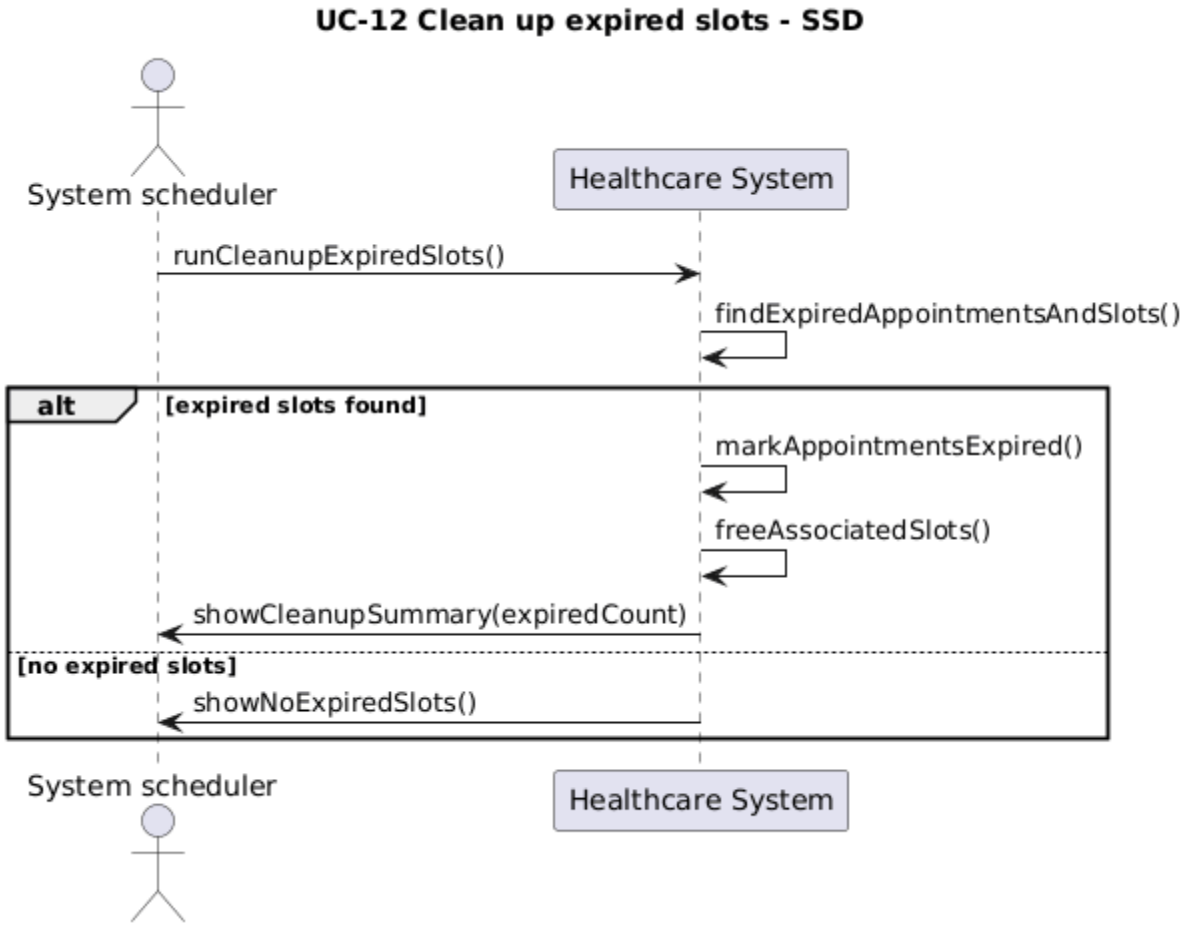
SSD – UC-10 Generate weekly report schedule



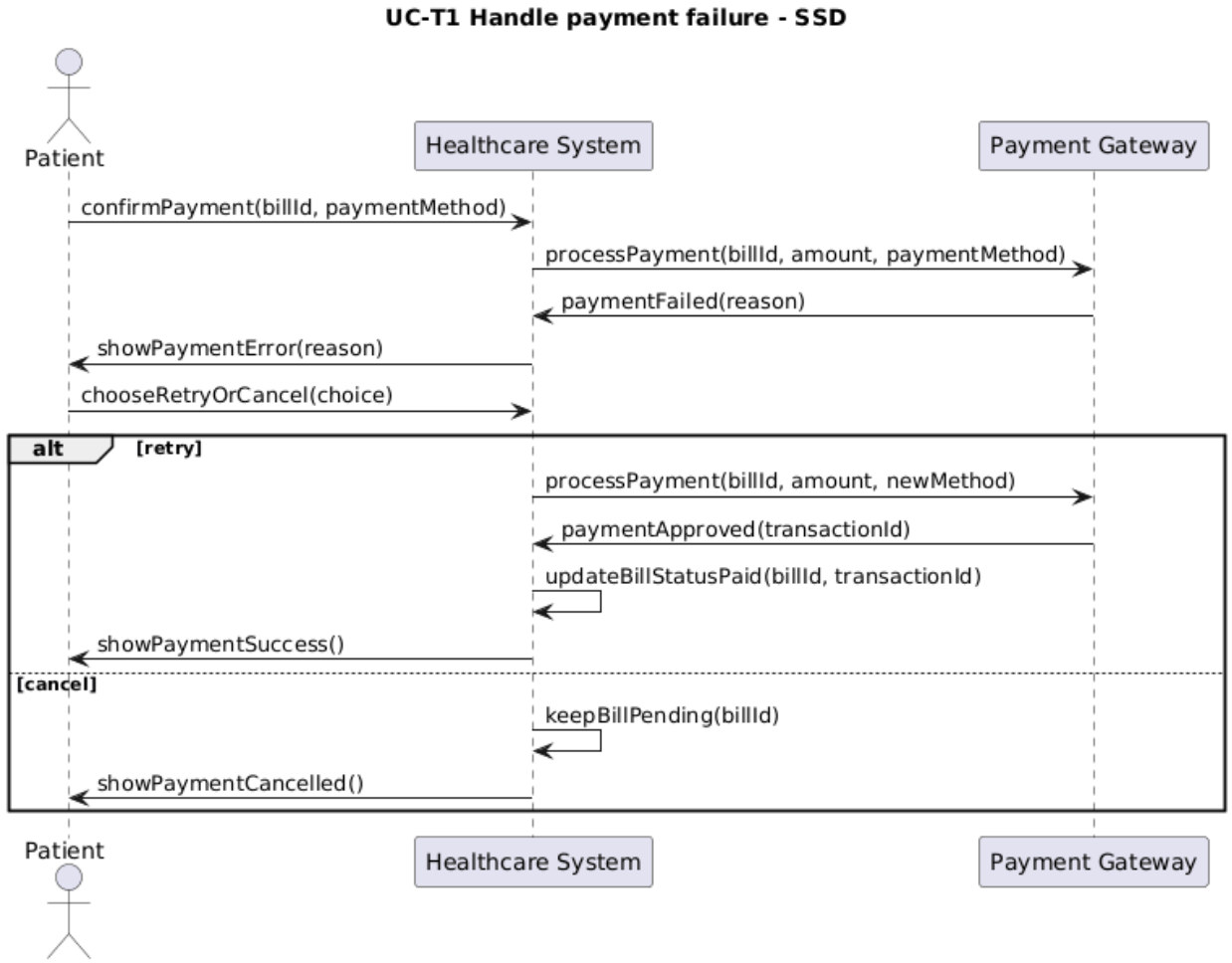
SSD – UC-11 Generate monthly bill



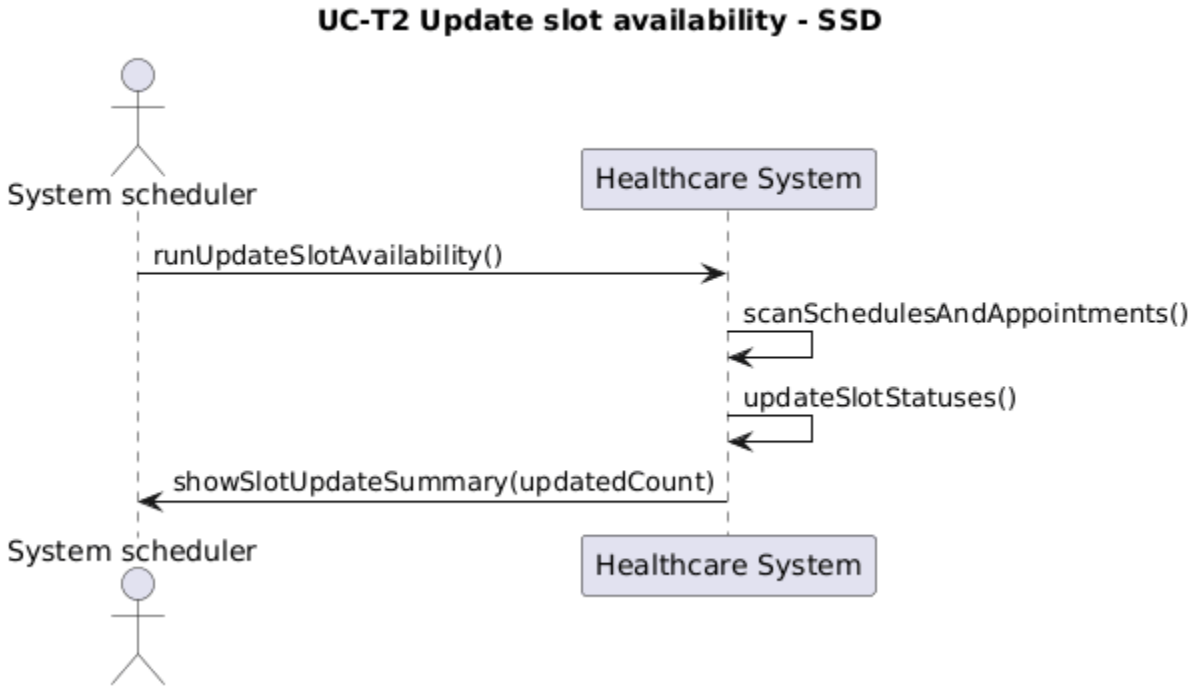
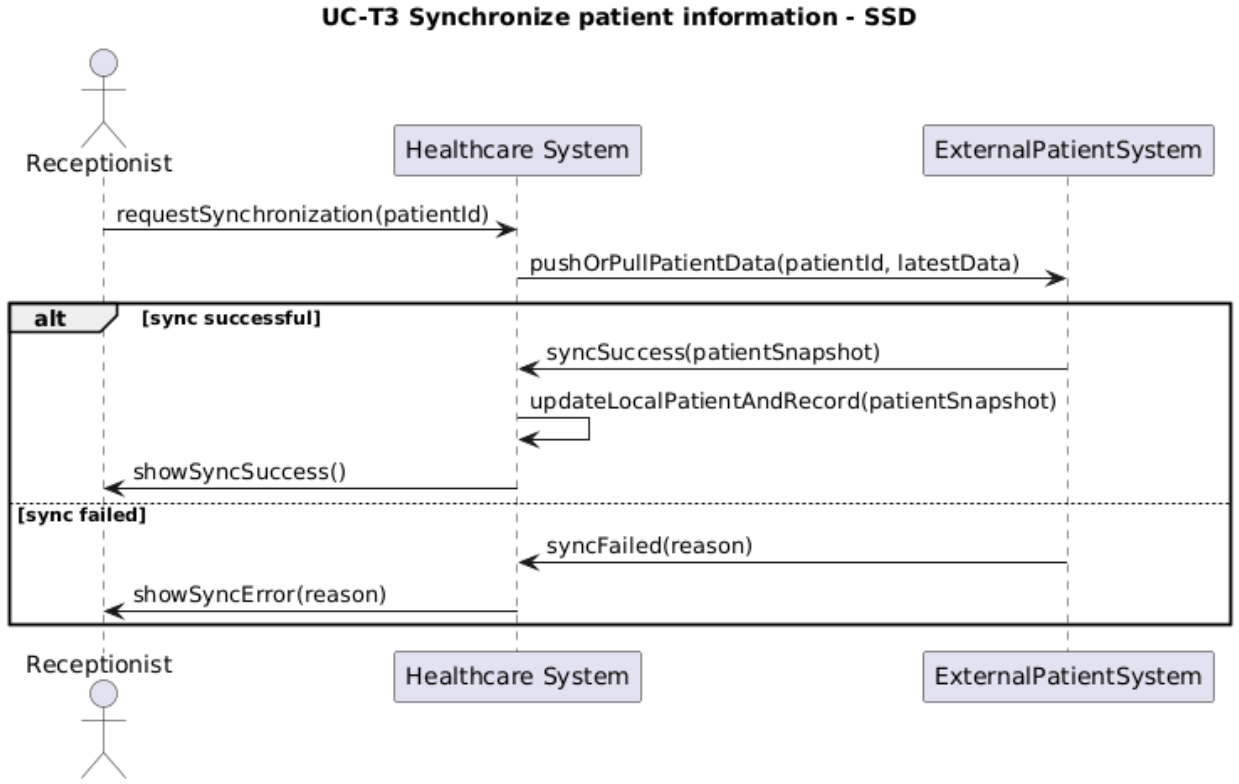
SSD – UC-12 Clean up expired slots



SSD – UC-T1 Handle payment failure



SSD – UC-T2 Update slot availability



SSD – UC-T3 Synchronize patient information